Shooting Stars From Halley's Comet

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Halley's Comet is far away more than 3 billion miles (4.8 billion kilometers) from Earth. But tiny pieces it left behind still light up our sky.

On October 21, photographer Tim McCollum captured three bright **Orionid meteors** above Toccoa, Georgia. The **Orionid** meteor shower happens each autumn, when Earth passes through dust and **debris** shed by **Halley's Comet**. The tiny pieces burn up in our air and become shooting stars.

McCollum took photos between 3 and 4 a.m. He used a Canon camera on a **star tracker** and took many 30-second pictures. Then he combined them into one image.

His picture also shows Orion's Belt, the Orion **Nebula** and the Rosette **Nebula**, with the Milky Way cutting across the sky.

The **Orionid** shower stays active until November 7. On clear nights, you may still spot one of Halley's shooting stars.

Word Treasure

Halley's Comet -- Halley's Comet is the faraway comet that left behind the tiny pieces seen as these shooting stars.

Orionid -- Orionid is the name of the meteor shower made from dust and debris shed by Halley's Comet.

debris -- Debris is small pieces left behind by something.

meteors -- Meteors are pieces that burn up in the air and look like shooting stars.

Nebula -- A nebula is a huge cloud of gas and dust in space.

star tracker -- A star tracker helps a camera follow the movement of stars.

Background you need

The Orionid meteor shower happens when Earth passes through dust and debris left by Halley's Comet.

As the tiny pieces burn in Earth's air, they make bright streaks called shooting stars.

A photographer can combine many long pictures to show several meteors in one sky image.

Break it down

WHO: Photographer Tim McCollum captured the meteors
WHAT: He photographed three bright Orionid meteors made from debris left by Halley's Comet
WHERE: Above Toccoa, Georgia
WHY: Earth passed through dust and debris shed by Halley's Comet, and the tiny pieces burned up as shooting stars

Why it matters

On a clear night, kids can look up and watch tiny pieces left by a faraway comet streak across the sky.

Test what you remember

1. How far is Halley's Comet from Earth?

- (A) Just past the Moon
- (B) More than 3 billion miles away
- (C) Closer than Mars
- (D) About one million miles away

2. What causes the Orionid meteor shower?

- (A) Earth passes through dust and debris shed by Halley's Comet
- (B) The Moon blocks the comet
- (C) Clouds reflect bright lights
- (D) Stars fall out of Orion's Belt

3. In which season does the Orionid shower happen each year?

- (A) Winter
- (B) Summer
- (C) Spring
- (D) Autumn

4. Who captured three bright Orionid meteors?

- (A) Ritu Raman
- (B) A Cambridge researcher
- (C) An astronaut
- (D) Tim McCollum

5. How did he make the final picture?

- (A) He took a photo from a rocket
- (B) He used one very quick photo
- (C) He drew the meteors by hand
- (D) He combined many 30-second pictures into one image

6. Which of these also appears in his picture?

- (A) A total eclipse
- (B) The International Space Station
- (C) The Rosette Nebula
- (D) A planet with rings

Answer key (try the quiz first!): 1: B 2: A 3: D 4: D 5: D 6: C

Questions to think about

MY THOUGHTS (at least 20 words):